

### Verizon ThingSpace (<https://thingspace.verizon.com/developer/>):

Verizon is the latest entrant in the IoT landscape. The Verizon Connectivity Management API allows users to add and activate devices, check their status, monitor their usage, and perform other device connectivity management tasks through a REST API. The API can be used to add connectivity management to anything, ranging from small apps to enterprise software systems, such as enterprise resource planning (ERP), supply chain management and customer service management. The API provides a secure, standards-compliant REST interface to the Web services at the Verizon M2M Data Center.

The ThingSpace platform tracks the entire life cycle of devices that are connected to the platform. It also provides a simulator to test the device APIs and once the application is developed, credentials can be changed to point to actual devices and get them involved in the platform.

### IoT alliances

Alliances help to set up a standard way of communicating and sharing data among all the stakeholders. They also provide a platform where these stakeholders can collaborate to validate and verify their offerings for adoption by others. Here are some of the IoT alliances that are prevalent.

**AllSeen Alliance (AllJoyn):** The first IoT standards group, AllSeen, was started by the non-profit Linux Foundation. It has more than 51 member organisations, including Microsoft, Qualcomm, LG, Sharp and Panasonic. AllSeen seeks to provide a secure, programmable software and services framework for applications that enable a connected home. Efforts are being made to standardise connectivity taking place through transport layers such as Wi-Fi, Wi-Fi-Direct, Ethernet, Powerline, Bluetooth LE, 6LoWPAN, ZigBee and Z-Wave. Interoperability is also a focus, with supported platforms including Android, iOS, Linux,

OpenWRT, Windows and OS X.

AllSeen is also committed to addressing security and privacy through its AllJoyn open source framework. An extension of the framework, the AllJoyn Gateway Agent, uses end-to-end encryption to keep communications secure. The Gateway Agent also uses privacy controls that let users decide which devices and applications have access to and from cloud services.

**Open Interconnect Consortium (OIC):** The OIC, led by Intel, Atmel, Broadcom, Dell and Samsung, is dedicated to defining requirements and ensuring interoperability of all devices in the IoT. Specifically, the OIC envisions a highway-like system of connectivity between IoT verticals. It recently launched IoTivity, an open source framework based on the Apache 2.0 licensing and governance model. The companies that make up the consortium also make security a top priority. The group is yet to address privacy clearly.

One differentiator for the OIC is that it wants to deliver a reference implementation of its IoT standards, rather than simply offering the standards themselves.

### The Thread Group (Google Nest):

Formed by Google's Nest Labs, the Thread Group includes more than 80 members, including Samsung, ARM Holdings, Silicon Labs and Freescale Semiconductor. The Group's goal is to encourage manufacturers of smart home devices to use the Thread standard for device communications through a network. Unlike other alliances that provide IoT platforms and interconnectivity of existing standards, Thread relies on a low-power radio protocol known as IPv6 over Low-power Wireless Personal Area Networks (6LoWPAN) as the base networking protocol. Thread sees this connectivity protocol as interoperable with the application layers provided by

the other alliances.

While the Group has said security has been built into the network and it promises to test this comprehensively on each product that it certifies, it is yet to document the data privacy plans.

**Apple HomeKit:** In the consumer IoT space, Apple wants to give third-party device makers approval under the 'Made for iPhone' certification process already used for iOS accessories. HomeKit will supply toolkits to developers to enable smart home integration for consumers. The HomeKit API features a common language designed to be interoperable with non-HomeKit devices that use protocols like ZigBee or Z-Wave.

HomeKit also includes security and privacy layers, such as end-to-end encryption between iOS devices and smart devices.

**IIC:** Founded by Intel, Cisco, AT&T, GE and IBM, the 150-member Industrial Internet Consortium (IIC) wants to accelerate Industrial IoT adoption while defining industry standards. The group's members are collaborating to develop connectivity standards, and the IIC has signed a strategic agreement with the OIC (Open Interconnect Consortium) to share information to streamline IoT device interoperability.

Additionally, the IIC is relying on its members, such as security platform provider CyberX and software giant SAP, to define standards for security and privacy.

There is a lot of excitement around IoT, and this is an effort to document the common frameworks and standards that are prevalent. There are new initiatives and activities that are happening in this space. If you need to implement an IoT strategy, pick a partner who understands the frameworks, sorts out the hype from true substance and has the expertise to implement the appropriate standards. 

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